Zhuoli Ouyang | Curriculum Vitae

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Personal Information

Date of Birth: 2004-10-15
Nationality: China
GPA: 3.94/4.00 (last year), 3.88/4.00 (total)
Languages: English, Chinese, Japanese

Research Interest

I am an undergraduate student with a strong foundation in mathematics and AI. My research interests span **generative models**, **reinforcement learning**, and **optimization**, with a focus on building more **efficient AI** systems. I am particularly excited about **embodied AI** and the challenge of enabling agents to perceive, reason, and act in complex environments. I aim to bridge principled mathematical thinking with practical system design, and look forward to contributing to research at the frontier of these areas.

Education

Southern University of Science and Technology–Second-year undergraduate — **Shenzhen**
Information Engineering, Dept. of Electronic & Electrical Engineering, China — 2024–Present
GPA: 3.88, Rank: **4/46**, Honor Class
Signals and Systems (97/100) Signal & Systems, Engineering Mathematics (95/100), Probability Theory and Mathematical Statistics (95/100)

Southern University of Science and Technology–First-year undergraduate — **Shenzhen**
Department of Mathematics, China — 2023-2024
GPA: 3.86, Rank: **94/1346** (overall in university), **Top 7**%, Fields Elite Class (Honor Class)
Algebraic Geometry (Graduate level), Advanced Functional Analysis (Graduate level), Analysis (97/100, Honor) Analysis, Advanced Algebra (97/100, Honor), C programming (95/100)

Tsinghua University High School — **Beijing**
High School Diploma, China — 2020–2023

School Affiliated to Tsinghua University — **Beijing**
Secondary Education, China — 2017–2020

Honors & Awards

2026: ALT 2026 Best Student Paper Award

2026: ISIT 2026 Strong Accept

2026: DATE 2026 Oral Paper (ACEMARL)

2024: SUSTech, Department of Electronics and Electrical Engineering-(Honor Class)

2023: SUSTech, Mathematics Department - Fields Elite Class (Top 1%, Honor Class)

Publications

RMNP: Row-Momentum Normalized Preconditioning for Scalable Matrix-Based Optimization — *ICML 2026*
Shenyang Deng, **Zhuoli Ouyang*** et al.* (*Equal contribution) — [arXiv] [ICML]

Understanding Generalization from Embedding Dimension and Distributional Convergence — *ICML 2026*
Junjie Yu, **Zhuoli Ouyang***, Haotian Deng*, Chen Wei, Wenxiao Ma et al.* (*Equal contribution) — [arXiv] [ICML]

Depth, Not Data: An Analysis of Hessian Spectral Bifurcation — *ISIT 2026*
Shenyang Deng, **Zhuoli Ouyang***, Boyao Liao* et al.* (*Equal contribution) — ***Strong Accept*** [arXiv]

Suspicious Alignment of SGD: A Fine-Grained Step Size Condition Analysis — *ALT 2026*
Shenyang Deng, Boyao Liao*, **Zhuoli Ouyang** et al.* (*Equal contribution) — [arXiv]

Doloris: Dual Conditional Diffusion Implicit Bridges with Sparsity Masking Strategy for Unpaired Single-Cell Perturbation Estimation — *ICLR 2026*
*Changxi Chi, Jun Xia, Yufei Huang, **Zhuoli Ouyang** et al.* — [OpenReview]

ACEMARL: Adaptive Clustering Enhanced Multi-Agent Reinforcement Learning for Analog Circuit Sizing *DATE 2026*
Han Wu, Haoning Jiang*, **Zhuoli Ouyang** et al.* *(* Equal contribution)* ***Oral Presentation***

FD-MAGRPO: Functionality-Driven Multi-Agent Group Relative Policy Optimization for Analog-LDO Sizing *AAAI 2026*
Haoning Jiang, Han Wu*, **Zhuoli Ouyang** et al.* *(* Equal contribution)* *[AAAI]*

Research & Study Experience

Dartmouth College **Hanover, NH**
Research Intern – Optimizers & Learning Theory 2025–Present
Advisor: Yaoqing Yang

Working on optimization algorithms and learning theory, contributing to theoretical analysis and algorithm design for machine learning.

Achievement: ICML 2026; ALT 2026 (**Best Student Paper Award**); ISIT 2026

AI for Scientific Simulation and Discovery Lab, Westlake University **Hangzhou**
Research Intern 2025–Present
Advisor: Tailing Wu

Working on generative AI methods (diffusion models, flow matching, stochastic optimization) for scientific simulation and discovery. Research spans AI for physical sciences—including fluid dynamics, plasma control, and embodied intelligence—as well as AI for life science, focusing on generative modeling of biological systems and evolutionary mechanisms.

Shanghai AI Laboratory / OpenDataLab **Shanghai**
Research Intern – Optimizer Benchmark & Survey 2025–Present
Advisors: Cheng Tan, Siyuan Li

Conducting a comprehensive benchmark and survey of optimization algorithms for large-scale AI, covering classical and modern optimizers across convergence, efficiency, and generalization.

HKUST (Guangzhou) **Guangzhou**
Research Intern – Theory-First Learning & Optimization for AI 2025–Present
Advisor: Yao Shu

Working on foundational optimization and learning theory for large-scale AI systems. Research spans black-box optimization (zeroth-order, Bayesian, bandit, RL), learning dynamics (generalization, stability, robustness), and their application to LLMs and intelligent agents.

Westlake CAIRI AI Lab, Westlake University **Hangzhou**
Research Intern – Generative Modeling for Biology (AI4Science) 2025–Present
Advisor: Changxi Chi

Working on generative modeling for biological systems, including diffusion models, flow matching, and bridge models for single-cell perturbation estimation and cell imaging.

Achievement: ICLR 2026

SUSTech **Shenzhen**
AI for EDA 2025
Advisor: Junmin Jiang

Applying Reinforcement Learning and AI algorithms to analog circuit design.

Achievement: AAAI 2026, DATE 2026, TCAD

Dept. of Math, Fields Elite - Honor Class & Graduate Courses **SUSTech**
Mathematics & Theoretical Physics (incl. FAU online) 2023–2025
Courses: Differential Manifolds, Algebraic Topology, Representation Theory, Abstract Algebra, Algebraic Geometry, Functional Analysis, Quantum Mechanics, General Relativity, Electromagnetism, etc.

See also: Algebraic Topology, Differential Forms in Algebraic Topology, Geometric Anatomy of Theoretical Physics

Other Skills

Programming: Python, C, C++, MATLAB, LaTeX, etc.
Software: PyTorch, PETSc, MATLAB, Fluent
Soccer: Second-class National Athlete in Football
Piano: Level 8, Central Conservatory of Music, Beijing
Hobbies: Exploring mathematics, Coding, Philosophy and History, Playing piano, Playing soccer